

ATKA, AK, USA

WMO: 700631

Lat: 52.220N

Lon: 174.206W

Elev: 17

StdP: 101.12

Time Zone: -10.00 (NAH)

Period: 06-19

WBAN: 25715

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.8	-11.2	1.4	-3.4	-9.8	1.6	-2.8	18.3	3.4	16.6	1.9	2.6	330	0.881

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	14.8	12.8	13.5	11.6	12.7	11.0	13.1	14.5	11.9	12.8	11.4	12.4	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.0	14.3	11.3	8.4	13.0	10.8	8.1	12.5	37.2	14.6	33.9	13.3	32.1	12.1	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	14.5	12.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	0.6	1.5	1.6	2.9	4.8	7.3	9.5	11.0	9.4	6.4	4.0	1.6
	DBStd	3.98	2.38	2.34	2.47	1.80	1.66	1.36	1.57	1.47	1.67	2.07	2.06	2.23
	HDD10.0	1861	291	237	259	213	163	81	28	5	30	112	180	262
	HDD18.3	4840	549	470	518	463	421	330	274	226	268	370	430	520
	CDD10.0	63	0	0	0	0	0	1	12	37	12	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.2	6.3	7.4	6.9	6.6	5.9	5.4	5.1	5.0	6.1	6.4	6.9	6.6
Precipitation	PrecAvg	1371	136	119	121	80	76	72	104	119	121	133	141	150
	PrecMax	1769	225	190	236	138	144	109	159	195	173	210	228	218
	PrecMin	1123	72	59	62	51	43	44	34	63	72	72	87	80
	PrecStd	128	35	29	39	19	20	16	30	32	29	36	32	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	8.1	8.0	9.0	11.2	13.0	16.1	17.5	14.5	12.6	9.4	7.6
		MCWB	4.4	5.2	4.9	4.9	7.3	10.2	13.7	14.8	12.1	10.7	7.8	5.6
	2%	DB	5.2	7.2	7.2	7.4	8.7	11.3	13.8	15.9	13.0	11.3	8.6	7.0
		MCWB	3.6	4.8	4.5	4.7	6.2	8.7	11.7	13.9	11.0	9.3	7.2	5.0
	5%	DB	4.1	6.0	6.1	6.3	7.7	10.0	12.6	13.9	12.5	10.2	7.7	6.0
		MCWB	2.7	4.2	3.8	3.9	5.5	8.1	10.9	12.1	10.7	8.3	6.3	4.2
	10%	DB	3.8	4.6	5.0	5.8	7.3	9.0	12.1	12.9	12.1	9.1	7.3	4.8
		MCWB	2.3	3.0	3.0	3.7	5.2	7.4	10.5	11.3	10.4	7.5	5.7	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	6.4	6.4	6.2	8.0	10.2	14.1	15.2	12.9	10.8	8.4	6.1
		MCDB	6.3	7.6	7.8	7.8	9.5	11.8	16.2	17.0	14.4	11.9	9.1	7.2
	2%	WB	4.0	5.2	5.0	5.2	6.8	8.9	12.2	13.9	11.7	9.6	7.8	5.1
		MCDB	5.3	6.5	6.6	6.9	7.8	10.3	13.4	15.5	12.7	10.9	8.3	6.7
	5%	WB	3.4	4.2	4.2	4.6	5.8	8.1	10.9	12.6	10.8	8.4	6.8	4.4
		MCDB	4.6	5.9	5.9	6.3	7.3	9.1	12.3	13.6	12.1	9.6	7.7	6.0
	10%	WB	2.8	3.3	3.3	3.8	5.2	7.7	10.2	11.8	10.2	7.8	5.8	3.5
		MCDB	3.9	4.6	5.0	5.5	7.0	8.6	11.5	12.6	11.5	8.8	7.1	4.8
Mean Daily Temperature Range	5% DB	MDBR	3.5	3.4	3.7	3.7	3.6	3.4	3.7	3.9	3.6	3.7	3.7	3.9
		MCDBR	3.5	4.6	4.8	4.8	4.5	5.4	5.5	5.9	4.7	4.3	4.4	4.7
		MCWBR	3.4	4.8	4.2	4.0	3.3	3.4	3.9	4.5	3.7	3.7	4.3	4.1
	5% WB	MCDBR	3.3	4.4	4.7	4.6	3.9	4.3	5.1	5.5	4.2	3.8	4.4	4.1
		MCWBR	3.4	4.6	4.4	4.1	3.2	3.2	3.8	4.5	3.6	3.7	4.9	4.1
Clear-Sky Solar Irradiance	taub	0.311	0.340	0.364	0.417	0.413	0.415	0.417	0.401	0.386	0.356	0.346	0.301	
	taud	2.327	2.282	2.262	2.108	2.188	2.262	2.316	2.368	2.314	2.297	2.178	2.308	
	Ebn at Noon	679	762	819	813	838	840	831	825	788	730	600	620	
	Edh at Noon	62	87	107	141	137	128	120	108	101	84	71	54	
All-Sky Solar Radiation	RadAvg	0.67	1.29	2.42	3.62	3.96	3.95	3.75	3.28	2.73	1.78	0.85	0.55	
	RadStd	0.07	0.14	0.19	0.34	0.32	0.38	0.27	0.28	0.19	0.17	0.08	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page